

EX.NO:9 STRING OPERATIONS USING ARRAYLIST

```
PROGRAM:
import java.util.ArrayList;
import java.util.Scanner;
public class StringMenu {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<String> list = new ArrayList<>();
        int choice;
        do {
            System.out.println("\n--- STRING OPERATIONS ---");
            System.out.println("1. Append");
            System.out.println("2. Insert");
            System.out.println("3. Search");
            System.out.println("4. Display Starting Letter");
            System.out.println("5. Display All");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Enter string: ");
                    list.add(sc.nextLine());
                    System.out.println("String added successfully.");
                    break;
                case 2:
                    System.out.print("Enter index: ");
                    int index = sc.nextInt();
                    sc.nextLine();
                    System.out.print("Enter string: ");
                    String value = sc.nextLine();
                    if (index >= 0 && index <= list.size()) {
                        list.add(index, value);
                        System.out.println("String inserted successfully.");
                    } else {
                        System.out.println("Invalid index.");
                    }
                    break;
                case 3:
                    System.out.print("Enter string to search: ");
                    String search = sc.nextLine();

                    if (list.contains(search))
                        System.out.println("String found.");
                    else
                        System.out.println("String not found.");
                    break;
                case 4:
                    System.out.print("Enter starting letter: ");
                    char ch = sc.nextLine().charAt(0);

                    System.out.println("Matching strings:");
                    for (String s : list) {
                        if (Character.toLowerCase(s.charAt(0)) ==
                            Character.toLowerCase(ch)) {
                            System.out.println(s);
                        }
                    }
                    break;
                case 5:
                    System.out.println("ArrayList: " + list);
                    break;
                case 6:
                    System.out.println("Program terminated.");
                    break;
                default:
```

```
System.out.println("Invalid choice.");
}
} while (choice != 6);
sc.close();
}
}
```

OUTPUT:

```
--- STRING OPERATIONS ---
1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit
Enter your choice: 1
Enter string: Apple
String added successfully.
```

```
--- STRING OPERATIONS ---
1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit
Enter your choice: 1
Enter string: Banana
String added successfully.
```

```
--- STRING OPERATIONS ---
1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit
Enter your choice: 2
Enter index: 1
Enter string: Orange
String inserted successfully.
```

--- STRING OPERATIONS ---

1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit

Enter your choice: 5

ArrayList: [Apple, Orange, Banana]

--- STRING OPERATIONS ---

1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit

Enter your choice: 3

Enter string to search: Banana

String found.

--- STRING OPERATIONS ---

1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit

Enter your choice: 4

Enter starting letter: A

Matching strings:

Apple

--- STRING OPERATIONS ---

1. Append
2. Insert
3. Search
4. Display Starting Letter
5. Display All
6. Exit

Enter your choice: 6

Program terminated.